



Insight

A strong sociotechnical perspective on digitalisation of and by projects: theoretically pluralistic, technologically agnostic and beyond techno-optimism

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ABSTRACT

Digitalisation plays an increasingly important role in projects. This insight note introduces how sociotechnical thinking can help with theorising, conceptualising and investigating digitalisation phenomena. It puts forward a strong sociotechnical perspective whereby the sociotechnical theory is seen through a theoretically pluralistic perspective used as a broad paradigm including other relevant theories that conceptualise the interactions or the entanglement of the social and technical parts. Additionally, this sociotechnical perspective emphasises technological agnosticism as long as the technological affordances are acknowledged to describe people's action possibilities. Through a mapping of the relevant theories to sociotechnical thinking, this insight note organises their ontological emphasis, potential empirical foci and units of analysis to understand various phenomena. Finally, through a critical view, the insight moves beyond technological determinism and techno-optimism to propose theoretical instruments distinguishing projects as objects (digitalisation of projects) and agents of digitalisation (digitalisation by projects), considering the latter as a long-term evolutionary process of system-level change brought forward by projects.

1. Introduction

Last year we launched the International Journal of Project Management (IJPM) call for papers on “Digitalisation of and by projects: A sociotechnical perspective on the consequences for project work” (Papadonikolaki et al., 2025a). After the submission, review, and publication process of this Special Collection was underway, we deemed it important to take stock of current theoretical developments and support the next stage of the collection. The motivation for this topic was grounded in the premise that, in the world of projects, digitalisation is not merely a technical concern but is inherently intertwined with social processes. Therefore, in the world of projects, digitalisation becomes primarily human centric (Papadonikolaki et al., 2025b). From this perspective, a sociotechnical lens is essential for advancing both project studies and our understanding of the digitalisation of and by projects. To support the further development of this premise, we therefore ask: What does a sociotechnical perspective on the digitalisation of and by projects entail? What do we really mean by sociotechnical? And what makes a sociotechnical perspective strong? Where to apply it and how can we evaluate such a sociotechnical standpoint?

In understanding the process of digitalisation and the variety of

relevant phenomena that are reshaping project studies, our sociotechnical perspective remains **technologically agnostic**. This means that any technological phenomenon is welcome as long as the impact on and contribution to projects, programmes, project portfolios and their management is strong. Digital technologies, such as Artificial Intelligence (AI), the Internet of Things, blockchain, big data, fundamentally influence how organisations operate and deliver projects (Badi, 2024; Whyte, 2019) and transform existing products, services, operations and work (Liu et al., 2024). Additionally, it is important when positioning the studied phenomena across the sociotechnical perspective to acknowledge their affordances. Gibson (1986) originally conceptualised affordances in ecological psychology as possibilities for action that the environment offers to humans, emphasising the actor-related nature of what is perceived as possible to do. Thus, affordances do not reside solely in an object but emerge in relation to the people perceiving and engaging with it. The concept of technological affordances originates from sociology (Hutchby, 2001) but has been developed in organisational research to describe the action possibilities that emerge through the interaction between actors and technologies (Leonardi, 2011). We hope that clear descriptions of the studied phenomena, explain **affordance-specific accounts** of what the technology means for whom and

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offer practical relevance methodological rigor and diversity (Unterhitzenberger et al., 2024) and subsequently well-defined theoretical contributions.

2. Importance of the sociotechnical perspective in studying projects

Project studies have increasingly embraced social theories. Social theories have long helped scholars explore and understand projects beyond their technical and instrumental dimensions. Floricel et al. (2014), for example, combined five dimensions of project practices—work and effort, materiality, agency and creativity, knowledge and interests, and power—with three fundamental social theories. They drew on Activity Theory, Actor-Network Theory and Structuration Theory to demonstrate how social theories provide valuable conceptual lenses for understanding contemporary project phenomena (Floricel et al., 2014). Their work represented an important step towards recognising projects as socially embedded and dynamically constituted systems unlike purely rational or technical endeavours.

Drawing upon Floricel et al. (2014), we further argue that a sociotechnical perspective can further our understanding of contemporary project phenomena amidst digitalisation. Digitalisation, however, changes the nature of project work, because technology is no longer simply a background artefact but an active component of organising and project organising. Digital technologies increasingly shift from nice-to-have to must-have features in projects. Therefore, theories that explicitly conceptualise the relationship between the social and the technical components become increasingly necessary in project studies. This was our main motivation for how the sociotechnical thinking originated.

The origins of sociotechnical thinking can be traced to the seminal work of Trist and Bamforth (1951) in the context of the coal mining industry. With the industrialisation waves of the society, the role of technology and the faith in it grew stronger. At times this was at the expense of the human factors and the social elements of work with technology. This was the need for the emergence of the sociotechnical approach where the social and the technical aspects are conceptualised as sub-systems of a higher-order system. To this end, sociotechnical theory emerged to counterbalance the dominant technocratic models that were technologically deterministic, ignoring human factors (Trist, 1981). Apart from the social and technical sub-systems, their sociotechnical interaction becomes central within an open system boundary, acknowledging their evolution and responsiveness to any changing conditions in the environment. Sociotechnical theory built on the two principles of sociotechnical interaction and joint optimisation, but like every theory it evolved too. At a later stage joint optimisation became less dominant due to the fact that the theoretical context evolved from heavy machinery and manufacturing to office-based work in services and large IT projects (Davis et al., 2014).

A sociotechnical perspective thus, suggests that any social system, as projects are often conceptualised, can only be understood and improved if both the social and technical aspects are considered together as interdependent parts of this complex system (Greenhalgh & Stones, 2010). Rather than viewing digital technologies as neutral tools that simply support existing work practices, this perspective recognises that digital technologies both shape and are shaped by the people, organisations, institutions and broader contexts in which they are embedded. For example, the implementation of AI in projects is not solely determined by the capabilities of AI algorithms, but also by project governance (Dacre, 2026), leadership, team capabilities (Shin, 2026), stakeholder interactions, organisational routines (Armanios & Tucci, 2025) and institutional norms. Likewise, these social processes are transformed by the introduction of AI, creating new ways of working (Barcaui & Monat, 2023), collaborating (Pei et al., 2025) and making decisions. A sociotechnical perspective therefore moves beyond technological determinism and social determinism, instead emphasising the

continuous co-evolution of social and technical elements and the emergent outcomes that arise from their interaction.

Building on Floricel et al. (2014), we argue that the rapid introduction of digitalisation in projects and project ecosystems calls for revisiting and extending the theoretical repertoire available to project scholars. Rather than proposing another single theory, we suggest that the sociotechnical perspective represents a broader family of theories, ranging beyond the original sociotechnical theory by Trist and Bamforth (1951), united by a common concern in understanding how social and technical elements jointly shape project outcomes and project missions. Therefore, apart from technologically agnostic, our sociotechnical perspective is also **theoretically pluralistic** but standing on appropriate theoretical grounding.

3. Theoretical perspectives adopting sociotechnical thinking

To this end, three broad families of theories emerge each with their different ontological assumptions about the relationship between the social and the technical aspects. Some are mainstream social theories, other cultural-historical and others organisational theories.

3.1. System-oriented sociotechnical perspectives

The first group of sociotechnical theories, already discussed, adopts a broader systems perspective, conceptualising projects as embedded within larger organisational, industrial and societal transformations. This relates to general systems theory and open systems theory (Von Bertalanffy, 1950). Instead of focusing primarily on situated interactions, these theories can explain how digitalisation unfolds across multiple levels of analysis and how projects both shape and are shaped by wider sociotechnical systems.

The classical Sociotechnical Systems Theory (Trist & Bamforth, 1951), Social Cognitive Theory (Bandura, 1986) and Social Network Theory (Granovetter, 1973) offer complementary explanations of how work systems, individual cognition and social relationships jointly influence the adoption and diffusion of digital technologies within projects. For instance, the social cognitive theory examines how new behaviours spread through society via social networks through psychosocial factors that influence this diffusion process (Badi, 2024). In digitally-enabled projects, social cognitive theory can show how psychological safety in teams spreads through multi-level learning and flat leadership dynamics (Papadonikolaki & Morgan, 2025).

Next, the Multi-Level Perspective (MLP) (Geels, 2011) conceptualises digitalisation as part of transitions across niche innovations, industry regimes and broader sociotechnical landscapes. Applying this perspective, Papachristos et al. (2024) demonstrated how megaprojects can accelerate digital change across the construction industry regime. At an even broader societal level, digitalisation has been associated with fundamental transformations of work itself, including debates on the emergence of a workless society and the role of projects in enabling transitions beyond existing industrial regimes (Frey & Osborne, 2017).

3.2. Actor- and process-oriented sociotechnical perspectives

The second group conceptualises digitalisation as an ongoing process emerging through the interaction between people, technologies and work practices. Activity Theory (Engeström, 1987), Actor-Network Theory (Latour, 2007) and Structuration Theory (Giddens, 1984) all reject deterministic views of technology, although they differ in their underlying ontological assumptions. Beyond treating technologies as fixed artefacts with predetermined effects, these theories emphasise that digital technologies acquire meaning and influence through their interactions with actors, practices and organisational contexts. In this sense, digitalisation is understood as a dynamic and socially situated process.

Using Activity Theory, Zomer et al. (2021) showed how the

implementation of digital technologies challenges existing activity systems and how socio-historical practices are re-enacted during the institutionalisation of new digital ways of working. Similarly, adopting an Actor-Network Theory perspective, [Linderoth \(2010\)](#) demonstrated how Building Information Modelling (BIM) redefines actors' roles and relationships by creating aligned and misaligned networks of human and non-human actors. [Harty and Whyte \(2010\)](#) further highlighted how actions are directed towards both material and digital objects within ecologies of practice. From a Structuration Theory perspective, [Papadonikolaki et al. \(2019\)](#) moved beyond viewing Building Information Modelling (BIM) as merely a technological tool and instead illustrated how project actors iteratively reshape structures and develop new ways of collaborating and delivering project work.

3.3. Complementary perspectives on sociotechnical phenomena

Beyond these core sociotechnical traditions, several complementary theoretical perspectives can enrich our understanding of the digitalisation of and by projects. Institutional Theory, for example, explains how organisational rules, norms and legitimacy shape digital transformation processes ([Korotkova et al., 2024b](#)), while Boundary Object theory provides insights into how digital technologies facilitate or challenge collaboration across organisational and disciplinary boundaries ([Korotkova et al., 2024a](#); [Papadonikolaki et al., 2019](#)). Boundary Object theory ([Star & Griesemer, 1989](#)) complements sociotechnical perspectives by explaining how digital technologies reshape organisational, disciplinary and knowledge boundaries, and how these changing boundaries influence collaboration and learning across project ecosystems.

Lastly, sociomateriality ([Orlikowski, 2007](#)) can be understood as an 'extreme' sociotechnical perspective and becoming particularly influential in studies of digitalisation. The origins of sociomateriality can be found at structuration theory ([Giddens, 1984](#)), whereby [Orlikowski \(1992\)](#) conceptualised the duality of technology arguing that technology is both shaped by humans and subsequently shaping their actions. Later, through a technology-in-practice perspective, [Orlikowski \(2000\)](#) argued that technological structures are enacted through repeated use rather than residing permanently within technological artefacts. In connecting sociomateriality with sociotechnical thinking, [Leonardi \(2012\)](#) sees sociomaterial practices as the 'technical subsystem' of sociotechnical systems. Indeed, sociomateriality encompasses different ontological positions, with [Jones \(2014\)](#) distinguishing between 'weak sociomateriality' (also mentioned as socio-materiality) with a more realist position, in which the social and material remain analytically distinguishable but intertwined and a 'strong sociomateriality' that considers them constitutively inseparable. Unlike traditional sociotechnical perspectives that distinguish between social and technical elements interacting while emphasising their interdependence, strong sociomateriality rejects the analytical separation of the social and material and assumes that the social and the material are inherently inseparable, entangled and mutually constitutive ([Orlikowski, 2007](#)), adopting a more realist ontology. Inspired by this perspective, [Morgan \(2019\)](#) demonstrated how digital technologies and organisational arrangements are co-constituted across user, organisational and institutional levels. Similarly, ([Pei et al., 2025](#)) employed sociomaterial ideas to explain how shared affordances emerge through the intertwining of technical and political processes in projects, revealing the dual role of digital technologies as both instrumental boundary objects and symbolic artefacts that confer legitimacy.

4. Empirical foci and units of analysis

Having introduced the principal theoretical perspectives, the next question becomes how they differ in terms of the empirical phenomena they explain. Drawing on [Whyte \(2019\)](#), digitalisation creates new properties of project information through four digital information

characteristics: e.g., (a) shareability, (b) remote accessibility, (c) searchability and (d) updatability, transforming project delivery. These properties generate different sociotechnical phenomena such as collaboration, coordination, knowledge integration, governance and adaptation and other empirical foci. And naturally, different sociotechnical theories and perspectives are used to explain different phenomena.

Although the sociotechnical thinking started from the context of a primary work system or organisational unit as the main units of analysis, other units of analysis were also considered early ([Trist, 1981](#)). Because of the acknowledgement of the open system boundary, the macrosocial unit of analysis became prominent too ([Trist, 1981](#)). This allows for studying the transformation of the sociotechnical interactions in evolutionary ([Geels, 2005](#); [Geels, 2004](#)) and long-term phenomena. Similarly, the other theoretical perspectives have too a primary empirical focus, or unit of analysis and a more expanded application area. As the above perspectives originate from social, organisational or context theories, [Table 1](#) below unpacks then further and illustrates how these various theoretical perspectives discussed above relate to different units of analysis important for the phenomena of digitalisation of and by projects.

5. Mapping the theoretical landscape

This Special Collection calls for papers on digitalisation, encompassing the impacts of digital technologies in changing project workflows and processes ([Papadonikolaki et al., 2025a](#)). We distinguish between **digitalisation of projects** and **digitalisation by projects**, the latter is related to digital transformation as a long-term evolutionary process of system-level change, and their consequences on project work. Together, these complementary perspectives recognise that projects both shape and are shaped by digitalisation, consistent with structuration ideas. They invite research that examines not only how digital technologies transform project organising but also how projects themselves become vehicles through which system-level sociotechnical change is initiated, negotiated and institutionalised across organisations, industries and society. Projects become both objects and agents of digitalisation.

First, digitalisation of projects continuously gives rise to new practices and processes in which project actors produce knowledge, products, artefacts, services to eventually deliver projects in innovative ways using digital technologies ([Whyte et al., 2016](#); [Papadonikolaki et al., 2022](#)). Second, digitalisation by projects drives digital change of permanent organisations by projects, programmes and portfolios as enablers of sociotechnical transformation ([Papachristos et al., 2024](#); [Papadonikolaki et al., 2023](#)), positioning projects not only as temporary organisations but also as mechanisms through which sociotechnical transitions are initiated, scaled and institutionalised. A previous editorial identified in the sister journal of International Journal of Project Management, Project Leadership and Society, showed that more research on the leading projects for digital transformation of companies and societies beyond the core project team and their outputs, such as organisations, communities and the society, is clearly required ([Papadonikolaki et al., 2025b](#)). [Table 2](#) provides some examples of how the aforesaid sociotechnical perspectives can contribute to both directions.

6. Conclusion

Through the integration of digital technologies, organisations undergo drastic changes in their work and project delivery ([Whyte, 2019](#); [Yang & Zhao, 2025](#)). This drastic impact of digitalisation affects the projects, organisations as well as their wider environment through novel ways the social and technical components interact and co-create value. But a truly sociotechnical perspective on digitalisation requires moving **beyond techno-optimistic** accounts of what digital technologies can enable and additionally examining the unintended consequences. Much

Table 1
Different theoretical perspectives and units of analysis related to digitalisation of and by projects.

Theoretical perspective	Theory and key reference	Ontological emphasis	Empirical foci and units of analysis (from individual to industry level)					Phenomena of digitalisation of and by projects
			Individual	Team/Activity system	Project/Network	Multi-projects/Ecosystem	Industry/Sociotechnical transition	
System-oriented	Sociotechnical Systems Theory (Trist & Bamforth, 1951)	Interdependence and joint optimisation of social and technical subsystems	+	[X]	[X]	++	+	Design and adaptation of work systems in which social and technical elements are jointly considered
	Social Network Theory (Granovetter, 1973)	Relational structures, ties and patterns of social interaction	++	[X]	++	[X]	+	Knowledge diffusion, collaboration, innovation networks and (digital) information flows
	Social Cognitive Theory (Bandura, 1986)	Cognition, behaviour and reciprocal interaction between individuals and their environment	[X]	++	+	-	-	Digital competences, learning, self-efficacy, leadership, behavioural change
	Multi-Level Perspective (Geels, 2011)	Nested interactions between niches, regimes and sociotechnical landscapes	-	-	+	[X]	[X]	(Digital) transitions across projects, industries and wider sociotechnical systems
Actor- and process-oriented	Structuration Theory (Giddens, 1984)	Recursive relationship between agency and social structure	++	[X]	[X]	++	+	Recursive interactions between actors, digital technologies and org. structures
	Activity Theory (Engeström, 1987)	Collective activity systems, mediation, contradictions and historically situated practice	+	[X]	[X]	+	-	Transformation of work practices, conflicts, learning, implementation of digital technologies
	Actor-Network Theory (Latour, 2007)	Relational ontology of heterogeneous human and non-human actors	++	++	[X]	[X]	++	Formation of sociotechnical networks, translation, alignment, agency of digital artefacts
Complementary ideas	Boundary Object theory (Star & Griesemer, 1989)	Boundaries between roles, organisations, professions, knowledge domains and practices	+	++	[X]	[X]	+	(Digital) boundary objects, coordination across organisations, domains and project phases, liminal roles, leadership
	Institutional Theory (Scott, 1995)	Rules, norms, meanings, legitimacy and institutional pressures	+	+	++	[X]	[X]	Standards, regulation, legitimacy, governance and institutionalisation of digital practices
	Sociomateriality (Orlikowski, 2007)	Constitutive entanglement and inseparability of the social and material	++	[X]	[X]	++	+	Mutual constitution of digital technologies, organising practices and institutional arrangements

Legend: [X] Main unit of analysis, ++ Common application, + Secondary application.

of the discourse surrounding digitalisation emphasises increased efficiency, integration, predictability and control. However, technologies do not automatically generate these outcomes, nor are their consequences necessarily beneficial or distributed equally, as they also depend on actors and their interactions with each other and the technology.

This critical position connects the digitalisation of and by projects with research on the dark side of projects. Locatelli et al. (2022) characterise the dark side as comprising sensitive, uncomfortable, ambiguous and often concealed phenomena that can harm individuals, organisations and society. Apart from project failure this may include unethical conduct, exploitation, discrimination, waste, greenwashing, political interests and other effects that may be suppressed or normalised within project environments (Aguilar Velasco & Wald, 2022). The dark side of digitalisation similarly encompasses more than the failure of a digital system to meet cost, time or performance expectations. It includes the work intensification (Papadonikolaki, 2026), algorithmic discrimination, job insecurity, digital divide (Dainty et al., 2017), 'digital-waging' and reduced wellbeing. A sociotechnical perspective should thus not overemphasise the technology affordances (Justesen & Plesner,

2024) of what technologies can do in projects but also what they cannot do, how they fail and what forms of tensions, exclusion, displacement, risk and inequality they may produce, phenomena that remain hidden behind narratives of innovation.

The profound effects of digitalisation through a variety of digital technologies in projects and their environment call for re-evaluating the nature of project management (Marnewick & Marnewick, 2022) and identifying an alternative toolset of theoretical perspectives to explain the novel phenomena of digitalisation of and by projects. In this insight note we explained how a sociotechnical perspective can create an umbrella paradigm including other theoretical perspectives to capture the multiplicity of relevant digitalisation phenomena and their impact. Through this **strong sociotechnical perspective**, that is not only social and not only technical, but a balanced combination of the two, acknowledging the interactions of social and technical and/or their entanglement is crucial in understanding digitalisation in project studies. We argue that a strong sociotechnical perspective on the digitalisation of and by projects has three characteristics: it is technologically agnostic, theoretically pluralistic and moves beyond techno-optimistic accounts to consider both what digital technologies

Table 2

Example alignment between sociotechnical perspectives and the foci of digitalisation of and by projects.

Theoretical perspective	Theory and key reference	Digitalisation of projects	Digitalisation by projects
System-oriented	Sociotechnical Systems Theory (Trist & Bamforth, 1951)	+	-
	Social Network Theory (Granovetter, 1973)	+	+
	Social Cognitive Theory (Bandura, 1986)	+	-
	Multi-Level Perspective (Geels, 2011)	-	+
Actor and process-oriented	Structuration Theory (Giddens, 1984)	+	-
	Activity Theory (Engeström, 1987)	+	-
	Actor-Network Theory (Latour, 2007)	+	+
Complementary ideas	Boundary Object theory (Star & Griesemer, 1989)	+	+
	Institutional Theory (Scott, 1995)	-	+
	Sociomateriality (Orlikowski, 2007)	+	+

enable and the limitations, tensions and unintended consequences they create. Naturally, the above list of theoretical perspectives is not exhaustive but can work as an open framework to be enriched by other theories that can support a strong sociotechnical perspective. The question is therefore no longer whether project scholars should adopt sociotechnical perspectives, but which sociotechnical perspective is most appropriate for explaining the increasing proliferation of digitalisation phenomena.

CRediT authorship contribution statement

Eleni Papadonikolaki: Conceptualization, Formal analysis, Project administration, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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